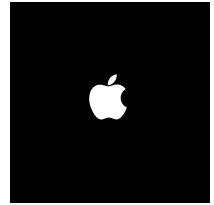


Interview Experience - Apple SDE Intern

About Apple

People at Apple don't just build products — they craft experiences that revolutionise industries. The diversity of our people and ideas drives innovation in everything we do. Imagine what you could do here! Join Apple, and help us leave the world better than we found it. At Apple, new ideas quickly become phenomenal products, services, and experiences. Every day, people achieve amazing things here. Apple is an Equal Employment Opportunity Employer, committed to inclusion, diversity, and affirmative action for minorities, women, veterans, and individuals with disabilities.



Apple's Expectation

Apple's IS&T team is looking for highly motivated and talented engineers to build the next generation of software services that power several mission critical applications. We are looking for extraordinary individuals with strong communication and interpersonal skills and ability to work effectively across multiple business and technical teams. As a part of this dynamic team, you will build cutting edge software service, one that is continually innovating & is proud of making a difference and help us build the Apple experience on a global scale. You should be a self starter, motivated, with exceptional interpersonal skills as well as a positive and flexible approach to problem

Eligibility Criteria

- cGPA 7 & above
- No Current Arrears
- CS / IT / AI&DS

Internship Details

- **Role:** Software Development Engineering Intern
- **Monthly Internship Stipend:** 105,000 INR
- **Relocation Benefits:** INR 35,100, less applicable taxes + Relocation Benefits (if applicable) economy class, round trip flight (if applicable) taxes + Relocation Benefits (if applicable)

Overview of Internship process

1. Online Assessment [Coderpad]
2. Resume Shortlisting [Implicit]
3. Technical Interview [Webex]
4. Techno-managerial Interview [Webex]

Online Assessment [90 mins]

MCQs: 10 Questions covered DSA, Networks, DBMS & OS, OOPS. Few of them are given below.

- Best Algorithm for finding longest palindromic substring
- Time complexity of Matrix Chain Multiplication
- Coffman conditions
- Hits during page replacement with LRU Cache
- SOLID principles
- Bandwidth-Delay product
- TCP congestion control

Problem Solving: Leetcode style Function completion

1. Restaurant Queue Management

A restaurant has n seats. An array *price* is given, where *price*[*i*] represents the amount the guest is willing to pay for being seated. An array *logs* is given, which represents the order of arrival of the guests. First entry of a *guestId* in *logs* refers to their arrival and the second entry refers to their departure. Constraints to be followed are,

- A guest can visit multiple times but they will be charged only once.
- A guest has to wait if all the seats are occupied
- If the guest loses patience waiting on the queue, he will leave the queue
- The scheduling has to follow FIFO. [not explicitly specified]

The objective is to find the profit of the restaurant.

Sample Input : $n = 2$, *price* = [2, 5, 8, 7], *logs* = [0, 2, 1, 3, 0, 3, 1, 2]

Output : 15

Source code: github.com/ZhaLabs/AppleInterview/Restaurant

2. Tennis Score

Simulate the scoring system of a Tennis game given the name of player1 *playerA* and player2 *playerB* and an array *series* where *series*[*i*] represents the name of the player who won the i^{th} serve. Constraints for defining the state of the game are

- A player wins if he has atleast 4 points & leads by atleast 2 points
- If both players have ≥ 4 points & tied, then its DEUCE
- If both player have ≥ 4 points & a there's a 1 point margin, then its Advantage for the player in Lead
- Points progresses as 0 -> 15 -> 30 -> 40

Further reference :

- **Article:** Tennis scoring system - Wikipedia
- **State diagram:** Tennis state diagram

Sample Input : *playerA* = "Jon Snow", *playerB* = "Tyrion Lanister"

series = ["Jon Snow", "Jon Snow", "Tyrion Lanister", "Jon Snow", "Tyrion Lanister", "Tyrion Lanister", "Tyrion Lanister", "Jon Snow", "Tyrion Lanister", "Tyrion Lanister"]

Output : "Tyrion Lanister WINS"

Source code: github.com/ZhaLabs/AppleInterview/TennisScorer

Resume Shortlisting

After the Online Assessment, candidates were shortlisted based on their score on the OE as well as their resume[implicit]. Shortlisted names were announced 15 days before the Technical interview. I wish to let you know that the flow of Technical interview & Techno-managerial interview and questions asked will depend on various factors including your resume, your recruiter, the team you have been chosen for[can be indentified from the linkedin profiles of your recruiter], the environment. In the upcoming segment i will sharing the experience from my perspective & leave to the readers to make inferences on the Interviewing process of Apple after reading all other interview experiences.

Technical Interview

The interview started with self introduction and then we proceeded to my project. I was asked the role I played in the project and later to explain the socket communication which we used in our project. Then I was asked core concepts from topics including Multi-threading, Context-switches, Normalisation, TCP handshake. Later I was asked my favorite subject. As I said DSA, I encounter questions on Trees, Time complexities and was asked to explain the working of *Heap* using Canvas on the Coderpad, operations & prove the time complexities of the ops & to implement the same. Later I was asked the preferred language to solve a problem. As I'm comfortable with multiple languages, I asked the interviewer to let me know the problem as I am problem-centric. He asked about the *TwoSum* problem. Once I explained him the two approaches, he asked me to implement the efficient one. I thought the interview was as we were already running out of the scheduled time. But asked me to solve another. I

was supposed to find the *longest common prefix between any pair of strings* in an array of strings. I gave him the brute force solution [$L * n^2$]. As he asked, I optimized it using sorting [$L * n \log n$]. He asked me to optimize it more. So I used Trie to make it efficient [$L * n$]

Source code:

- github.com/ZhaLabs/AppleInterview/MinHeap
- github.com/ZhaLabs/AppleInterview/TwoSum
- github.com/ZhaLabs/AppleInterview/LongestCommonPrefix

Within 30 mins I received the mail stating the schedule for the next round and I was supposed to confirm my availability ASAP.

Techno-Managerial Interview

As I was in a hurry, Interview asked me reason behind that. As I confessed about my examination, we went on sharing each others' college life. He asked my commitments later on the day even before asking self-introduction. I felt it had set a comfortable stage for open talks. Then I was asked the following questions.

- Tell me mistake/setback that you see in Apple
- Explain your project.
- What was your role in the project?
- Tell me about your interview experience
- Given the following attributes[Name[duplicates present], Contact[0..*], Courses[0..*]], design a relational schema
- What the challenges you faced during your project
- Explain the socket programming
- Explain an algorithm that you very comfortable with
- Recent breakthrough in the algorithm[MST] I explained
- What are you key strengths?
- Prove me your leadership with a scenario
- How you handled a time where your teammate was not able to cope up with your project's progress?

Even after all these question, there was 10 mins remaining in the scheduled time. I wished to utilize the time to prove myself. So I kept questioning him for more than 20 mins from the inferences I made from his LinkedIn profile. Results were announced after 4 from the last day of Interview process.

Personal Advice

- Use the self-intro part to set the perfect tone for the rest of the interview.
- Navigating the interviewer to your comfort zone was very hard when comparing to interviews I attended in other companies.
- My interviewers had 9 & 20 years of experience respectively. So they will find if you pretend to project yourself as expert in the topic.
- Recruiter were very experienced. So don't prove you explicitly. Just give them cues. They will scan you.
- During pair-coding think loudly i.e express your thought on the problem to the interview
- As said earlier, questions depends on the interviewers. When asked to solve problems, try to ask the problem first then let him know language of your choice. I'm repeating because, a friend of mine who very much equipped with the required skillset, encounter a question which I felt unfair. He chose C++ but was supposed to store objects of different type in same container and was supposed to do object destructuring.
- Smile
- Resources : Refer CSEA

Fingers Crossed!!!